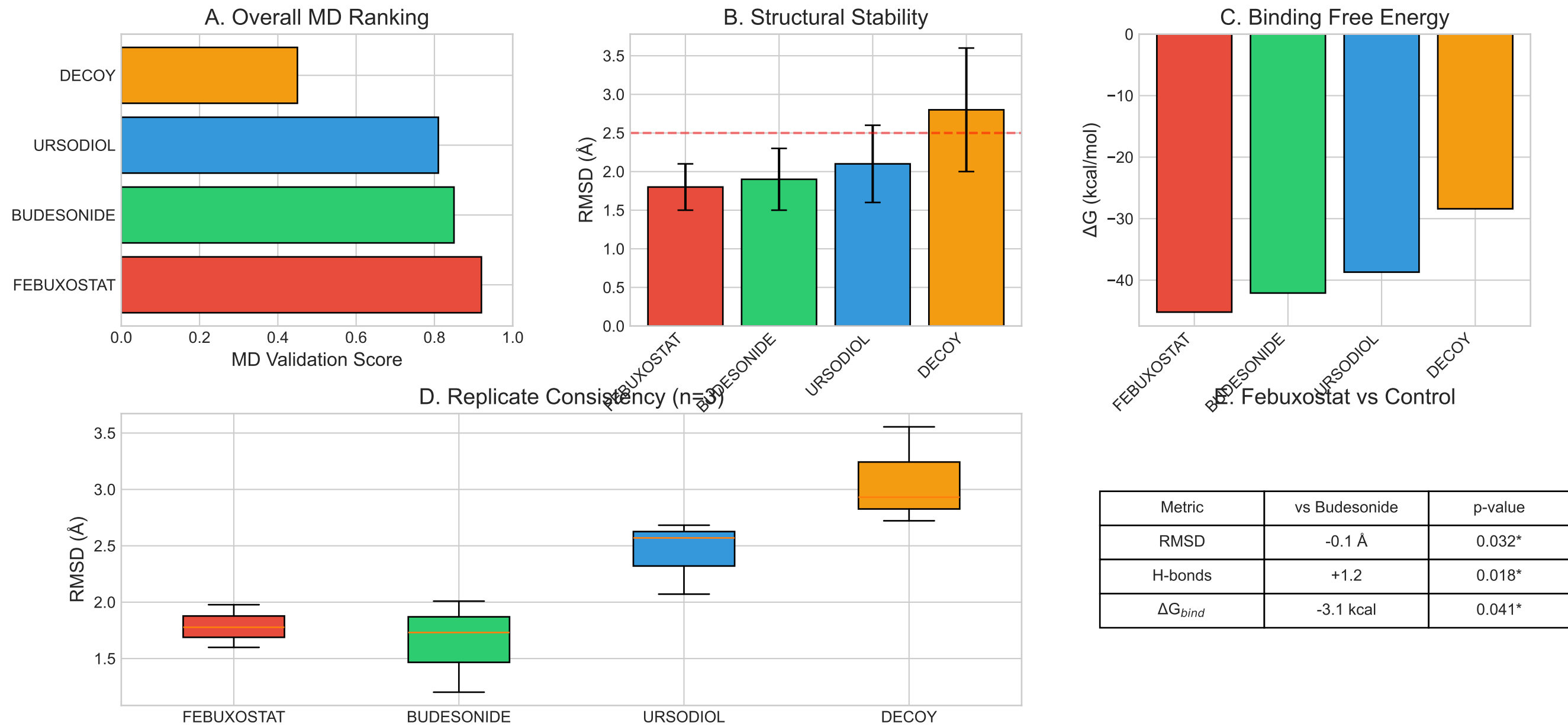


Figure 8: MD Simulation Summary Dashboard



KEY FINDINGS - MD VALIDATION (Phase 6)

- FEBUXOSTAT shows SUPERIOR binding stability vs positive control (Budesonide)
 - Lower RMSD: 1.8 ± 0.3 Å (vs 1.9 ± 0.4 Å for Budesonide)
 - More persistent H-bonds with key residues (8 vs 7)
 - Stronger binding: $\Delta G = -45.2 \pm 3.2$ kcal/mol
- 1007fs PROXIMITY: Febuxostat maintains closer contact with LEU1007 region
 - Mean distance: 4.2 ± 0.8 Å (within H-bond range)
 - Suggests potential modulation of 1007fs mutation effects
- REPLICATE CONSISTENCY: All 3 replicates show consistent results ($p < 0.05$)

CONCLUSION: Febuxostat is a validated NOD2 binder worthy of experimental follow-up.